



# Biochemical biomarkers for the toxicity induced by Traditional Chinese Medicine: A review update

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## ABSTRACT

**Ethnopharmacological relevance:** Traditional Chinese medicine (TCM) is widely used in China for disease treatment and has become a valuable resource for drug development due to its high efficacy and low risk of side-effects. However, growing toxicity reports has garnered significant global attention. A major challenge in addressing TCM-induced toxicity is lack of specific and sensitive biomarkers for diagnosing and predicting its toxicity. Identifying toxicological biomarkers reflecting TCM-induced toxicity is crucial for timely detection and intervention, and provides significant clues for elucidating the underlying toxic mechanism and key target.

**Aim of the study:** This article aims to summarize and classify some potential toxicological biomarkers for side-effects induced by TCM and its contained phytochemical ingredients.

**Methods:** The keywords “biomarkers”, “traditional Chinese medicine”, “Chinese herb”, “phytochemical ingredient”, “natural product”, “toxicity”, “hepatotoxicity”, “nephrotoxicity”, “cardiotoxicity” were used to collect relevant information from literature databases (including PubMed, Web of Science) up to October 2024.

**Results:** Research has indicated that more sensitive and specific biomarkers are needed for reflecting TCM's side-effects. PA-protein adducts and AA-DNA adducts could be served as diagnostic biomarkers for hepatotoxicity and nephrotoxicity induced by TCM containing PA and AA, respectively. Multiple miRNAs like miRNA-122-3p, miRNA-5099, and miRNA-21-3p, as well as some endogenous metabolites such as hypoxanthine, choline, and L-valine could be potential biomarkers associated with TCM-induced hepatotoxicity, nephrotoxicity, and cardiotoxicity.

**Conclusion:** In this review, different research demonstrates that DNA/protein-adducts, noncoding RNAs, endogenous metabolites and so on show the potential to be new early-warning biomarkers for TCM-induced toxicity with high specificity and sensitivity.

## 1. Introduction

Traditional Chinese medicine (TCM) is a prevalent treatment for

various diseases in China, valued for its therapeutic efficacy and minimal side-effects (Tong and Deng, 2020). However, increasing reports of adverse reactions induced by TCM stemming from improper use,

**Abbreviations:** AA, aristolochic acid; ADR, adverse drug reaction; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; BNP, brain natriuretic peptide; CFDA, China Food and Drug Administration; CHE, Cholinesterase; CYP, Cytochrome P450 proteins; DHPA, dehydro-pyrrolizidine alkaloid; DIKI, drug-induced kidney injury; DILI, drug-induced liver injury; DV, *Dysosma versipellis*; HSOS, hepatic sinusoidal obstruction syndrome; KIM-1, kidney injury molecule-1; lncRNA, long non-coding RNA; MCP-1, monocyte chemoattractant protein-1; miRNA, MicroRNA; PA, pyrrolizidine alkaloid; PMR, Polygonum multiflorum Radix; SQLE, Squalene synthase; TCM, Traditional Chinese Medicine; TNF $\alpha$ , tumor necrosis factor alpha; TSG, 2,3,5,4'-tetrahydroxyl-diphenylethylene; TSN, too-sendanin; VEGF, vascular endothelial growth factor.

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contamination, or inherent toxicity has raised significant concerns. These issues not only affect the safe and effective usage of TCM in clinic but also impede further drug development and international acceptance (Chan et al., 2015; Claire-Del and Espinosa-Cuevas, 2021; Teschke et al., 2021). As described in the annual report of national adverse drug reaction (ADR) monitoring by China Food and Drug Administration (CFDA), there were a total of 2.02 million cases related to ADR in China, of which TCM contributed to 12.8%. It was reported that the intake of TCM is usually in connection with cardiotoxicity, nephrotoxicity, neurotoxicity, and especially hepatotoxicity (Chen et al., 2011). Taking the hepatotoxicity of TCM as an example, a nationwide retrospective study in 2019 suggested that 26.81% of the cases were related to TCM in 25,927 patients diagnosed with drug-induced liver injury (DILI) (Shen et al., 2019; Li et al., 2022). The complexity of TCM, including the multi-component nature and diverse pharmacological interactions, contributes to the difficulty in diagnosing and managing its toxicity. Furthermore, the multifaceted and often delayed manifestations of TCM-induced toxicity, which may affect multiple organs, pose challenges for timely and accurate diagnosis, thereby limiting TCM's clinical application and global advancement. Hence, it is crucial to identify the key molecules in TCM-induced toxicity and develop proper diagnostic approaches.

Biochemical markers, which reflect changes in the biological functions of cells and organs, can indicate the presence, progression, or improvement of diseases (Schomaker et al., 2019). Since the 1990s, more and more researchers have devoted to exploring biomarkers due to their role in facilitating and accelerating the access to obtain the information about drug efficacy and safety (Cohen et al., 2015). In order to avoid the occurrence of side-effects as much as possible and ensure the safe use of TCM in clinic, it is of great significance to identify the potential injury as soon as possible before the occurrence of pathological damages induced by TCM. Therefore, the discovery of biomarkers for toxicity induced by TCM will be helpful to reduce their potential ADR and guide their clinical medication safely and effectively, and it will also provide key clues for its detoxification. However, the current key problem for toxicity induced by TCM is the shortage of specific, sensitive and accessible biomarkers for diagnosing and predicting its toxicity, and many classic biomarkers, such as alanine aminotransferase (ALT), aspartate amino transferase (AST), blood urea nitrogen (BUN) and so on, are all insensitive and nonspecific (Griffin et al., 2019).

In recent decades, many toxicological researchers focused on discovering biochemical markers with the characteristics of high specificity, high sensitivity, and good stability for toxicity induced by TCM. Among these studies, protein or DNA adducts formed by toxic substances have emerged as promising biomarkers for TCM-induced toxicity. For instance, dehydro-pyrrolizidine alkaloid (DHPA)-protein adducts have been successfully incorporated into clinical diagnostics for pyrrolizidine alkaloid (PA)-induced hepatic sinusoidal obstruction syndrome (HSOS), forming part of the Nanjing criteria (Zhuge et al., 2019). Additionally, non-coding RNAs, known for their high sensitivity, have been reported as potential biomarkers for TCM-induced toxicity. Furthermore, endogenous metabolites and proteins, including cytokines and enzymes, which not only reflect the occurrence of disease but also directly participate in its progression, hold great potential as biomarkers for monitoring TCM-induced toxicity. This review aims to comprehensively summarize the biomarkers identified to date for TCM-induced toxicity, providing valuable insights to ensure the safe and effective clinical application of TCM.

## 2. Classic biomarkers for toxicity induced by TCM

Classic biomarkers of toxicity typically refer to those indicators that have been widely used in clinical practice for decades to evaluate drug-induced toxicity or monitor disease status, and they have also been commonly utilized to reflect TCM-induced toxicity. As summarized in Table 1, classical biomarkers include but are not limited to, liver

**Table 1**

Classic biomarkers for toxicity induced by TCM.

| Type of toxicity | Classic biomarker                          | Object | Reference                                 |
|------------------|--------------------------------------------|--------|-------------------------------------------|
| Hepatotoxicity   | Alanine aminotransferase                   | Serum  | Gao et al. (2023)                         |
|                  | Aspartate aminotransferase                 | Serum  |                                           |
|                  | Alkaline phosphatase                       | Serum  |                                           |
|                  | Total bilirubin                            | Serum  |                                           |
| Cardiotoxicity   | Cardiac troponin T                         | Serum  | Wang et al. (2019)                        |
|                  | Cardiac troponin I                         | Serum  |                                           |
|                  | B-type natriuretic peptide                 | Serum  |                                           |
|                  | N-terminal pre-B-type natriuretic Titanium | Serum  |                                           |
| Neurotoxicity    | Neuro-specific enolase                     | Serum  | Korfias et al. (2006); Lian et al. (2023) |
|                  | S100 protein                               | Serum  |                                           |
| Nephrotoxicity   | Blood urea nitrogen                        | Serum  | Gao et al. (2023); Sun et al. (2011)      |
|                  | Creatinine                                 | Serum  |                                           |

enzymes like ALT and AST for assessing hepatotoxicity (Gao et al., 2023), and cardiac troponin and brain natriuretic peptide (BNP) for detecting cardiotoxicity caused by TCM (Wang et al., 2019). In addition, BUN and serum creatinine were utilized to assess nephrotoxicity induced by TCM (Sun et al., 2011). These biomarkers are mainly collected from serum or urine, and they have the characteristics of easy detection. However, due to the complexity of TCM components, these conventional biomarkers often struggle to accurately assess the specific toxicity characteristics of TCM. Moreover, they often exhibit limitations in terms of specificity and sensitivity, reducing their effectiveness in distinguishing TCM-induced toxicity from other forms of toxicity or underlying conditions. Thus, while these classical biomarkers remain valuable tools, their limitations highlight the need for more advanced biomarkers to reflect the unique toxicological effects of TCM.

With the advancement of medical research and experimental technologies, the exploration of biomarkers has reached new depths, leading to the discovery of novel biomarkers associated with toxicity induced by TCM. These emerging biomarkers show greater potential in accurately reflecting the toxicity characteristics of TCM.

## 3. Protein or DNA adducts as biomarkers for toxicity induced by TCM

Previous study has reported that drug-protein/DNA adducts were considered as an important feature of pharmacological or toxicological indicators, and they can be easily detected (Gan et al., 2016). Studies have already demonstrated that acetaminophen-protein adducts could be served as early biomarkers for monitoring acetaminophen-induced acute liver injury in clinic (Chiew et al., 2020; James et al., 2009). Meanwhile, DNA adducts have also been recognized as biomarkers for diseases (Hernandez-Castillo et al., 2020). Hence, protein or DNA adducts exhibit great potential to serve as biomarkers for toxicity induced by TCM.

### 3.1. Protein adducts as the biomarker

PA, a type of natural hepatotoxic alkaloid, is globally distributed in plants belong to Compositae, Boraginaceae, Leguminosae and Orchidaceae families, among which many plants are listed in China Pharmacopeia or commonly used as folk medicines, such as *Gynura japonica*, *Senecio scandens* Buch.-Ham., *Arnebia euchroma* (Royle) Johnst. et al. (Chojkier, 2003; Zhang et al., 2008; Luo et al., 2019). Clinical reports found that herbal medicines containing PA could induce acute liver toxicity with representative pathological symptoms including hepatomegaly, jaundice, and ascites, which is known as HSOS (DeLeve et al., 1999; Chojkier, 2003; Zhu et al., 2021a). Hepatic cytochrome P450 was discovered to catalyze PA into DHPA, which can further covalently bind to macromolecules like RNA, DNA or protein, causing toxicity (McLean, 1970; Fu et al., 2004; Li et al., 2011; Fu, 2017).

More and more reports have exhibited that DHPA-protein adducts hold significant promise as biomarkers for hepatotoxicity induced by PA (Lu et al., 2018). A highly significant positive correlation was found between serum ALT enzymatic activity and the formation of DHPA-protein adducts in liver injury induced by PA, indicating its potential application as a biomarker for hepatotoxicity induced by PA (Ma et al., 2018). Later, a DHPA-protein adduct was identified in the serum from a HSOS patient caused by the accidental ingestion of *Gynura japonica* (Tu-San-Qi) that is a folk herbal medicine containing hepatotoxic PA (Lin et al., 2011). Furthermore, a prospective study of forty patients, among whom 23 patients had PA-associated HSOS and 17 patients had liver injury caused by other drugs without PA, demonstrated that blood concentrations of DHPA-protein adducts ranged from 0.05 to 74.4 nM in PA-associated HSOS patients (Gao et al., 2015). This study demonstrated that the measurement of blood DHPA-protein adducts exhibited 100% sensitivity and 94.1% specificity in the diagnosis of PA-associated HSOS (Gao et al., 2015). Moreover, the levels of blood DHPA-protein adducts were found to be significantly correlated with the severity and clinical outcome of PA-related HSOS, because those patients who died or required liver transplantation within 6 months (categorized as the severe group) had significantly higher blood DHPA-protein adducts levels compared to those who fully recovered or developed chronic disease (categorized as the recovery/chronic disease group) (Gao et al., 2015). These clinical findings highlight the huge potential of blood DHPA-protein adducts as specific and sensitive biomarkers for assessing PA-induced hepatotoxicity. Furthermore, the detection of DHPA-protein adducts has already been integrated into clinical diagnostics for PA-induced HSOS, as outlined in the Nanjing criteria (Zhuge et al., 2019).

DHPA-protein adducts have also been found in experimental studies involving mice and rats intoxicated by herbal medicines containing PA. For example, a high level of DHPA-protein adducts was found in the blood and liver both in mice and rats after the administration of *Gynura japonica* (Yang et al., 2017a). Although there is still no report about the hepatotoxicity induced by other herbal medicines containing PA in clinic, the extract of *Crotalaria sessiliflora* was found to induce severe hepatic injury in rats and DHPA-protein adducts were also detected in liver tissues (Ma et al., 2021a). These findings suggest that DHPA-protein adducts could be utilized as specific biomarkers for monitoring liver injury induced by PA-containing herbal medicines.

Recently, a DHPA-hemoglobin adduct was identified in the blood from 43 patients with PA intoxication, and this adduct displayed obviously higher levels and longer persistence, suggesting its potential as a more reliable biomarker for diagnosing hepatotoxicity caused by PA-containing herbal medicines (Ma et al., 2021b). Meanwhile, another study manifested that there were four DHPA-amino acid adducts in urine samples from rats intoxicated with PA and patients who had ingested *Gynura japonica*, named pyrrole-7-cysteine, pyrrole-9-cysteine, pyrrole-9-histidine, and pyrrole-7-acetylcysteine (Zhu et al., 2021b). These four DHPA-amino acid adducts exhibited strong potential as non-invasive biomarkers for assisting in diagnosing hepatotoxicity induced by PA-containing herbal medicines.

Except PA, other toxic substances from toxic TCMs can also form protein adducts. Geniposide, a main active natural compound isolated from *Gardenia fructus*, can be hydrolyzed into genipin by intestinal microbiota  $\beta$ -glucosidase, and genipin was found to interact with L-lysine residue in the protein to form genipin-protein adducts, thereby inducing hepatic injury in rats (Li et al., 2019). Toosendanin (TSN) is the main hepatotoxic compound found in *Toosendan fructus*, which is marked to have little toxicity in Chinese Pharmacopeia. A previous study manifested that protein adducts of the reactive metabolites of TSN could be found in the liver and blood from rats, and those protein adducts displayed a good correlation with the severity of TSN-induced hepatotoxicity (Zhuo et al., 2021). Traditional herbal medicine *Dioscorea bulbifera* L. has been found to cause hepatic damage during clinical practice, and experimental results also confirmed its hepatotoxicity (Ma

et al., 2014). Diosbulbin-B, the primary hepatotoxic compound from *Dioscorea bulbifera* L., could be metabolized by hepatic cytochrome P450 to produce its metabolite that can further bind to lysine residues of protein and finally form protein adducts, which could be detected in the plasma and showed a positive relationship with doses of diosbulbin-B (Hu et al., 2022). However, whether genipin-protein adducts, TSN-protein adducts, or diosbulbin-B-protein adducts can be used as reliable biomarkers for their toxicity still requires more evidences from clinical investigation.

### 3.2. DNA adducts as the biomarker

DHPA-DNA adducts have also been discovered in the livers from rats given with a series of hepatotoxic PAs or dietary supplements (comfrey or coltsfoot root extract, comfrey compound oil) or *Tussilago farfara* L. (a medicinal herb) (Chou and Fu, 2006; He et al., 2017). Additionally, studies have shown that DHPA-DNA adducts exhibit long-term persistence in mice given with PA, indicating their huge potential as biomarkers for reflecting hepatotoxicity induced by PA (Zhu et al., 2017). A recent study further identified a good positive correlation between liver DHPA-DNA adducts and serum DHPA-protein adducts (Zhu et al., 2022). Since DHPA-protein adducts exhibit higher abundance in serum and are easier to be detected, the authors propose that it is more suitable than liver DHPA-DNA adducts as biomarkers for indicating PA intoxication.

Aristolochic acid (AA) is abundant in various TCMs, such as *Aristolochia manshuriensis* Kom., *Aristolochia fangchi* Y. C. Wu ex L. D. Chow et S. M. Hwang., *Aristolochia debilis* Sieb. et Zucc, and so on. AA was reported to induce severe acute renal injury and tubulointerstitial kidney disease (Yang et al., 2018; Jelakovic et al., 2019; Baudoux et al., 2022). AA could be converted into its metabolic active product aristolactam that covalently binds to purines in DNA to form aristolactam-DNA adducts, which can cause kidney injury and also exhibit strong carcinogenic activity (Stiborova et al., 2017; Anger et al., 2020). A clinical investigation (Jelakovic et al., 2012) manifested that aristolactam-DNA adducts were detected in renal cortex from 47 patients with nephrosis induced by AA. Moreover, aristolactam-DNA adducts displayed good sensitivity and stability, and they were resistant to global genomic nucleotide excision repair, so they have the huge potential to be reliable biomarkers for diagnosing nephropathy induced by AA-containing TCMs (Moriya et al., 2011; Jelakovic et al., 2012).

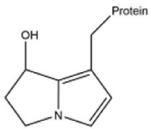
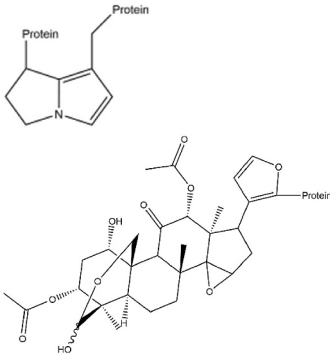
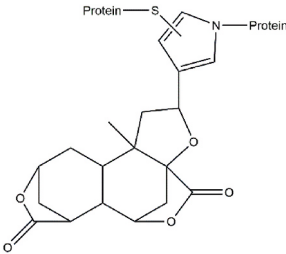
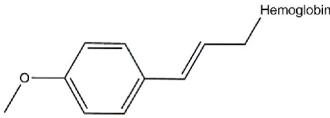
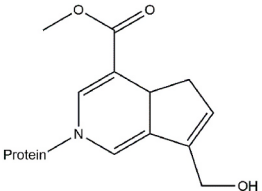
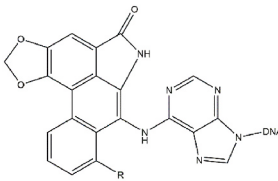
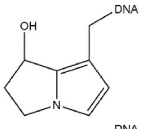
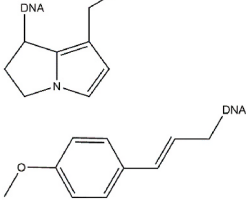
Estragole from *Ocimum basilicum* L. and *Illicium verum* Hook. f. have been reported to be metabolized by hepatic cytochrome P450, and its metabolites can form DNA-adducts that are considered potential biomarkers for reflecting their hepatotoxicity (Rietjens et al., 2005; Chen et al., 2011). Furthermore, DNA-adducts and hemoglobin-adduct were formed when estragole was incubated with hepatic microsomes (Bergau et al., 2021). However, the clinical application of those DNA- or hemoglobin-adducts requires further validation through more robust experimental and clinical evidences.

All those above studies imply that those protein- or DNA-adducts (Table 2) (Fig. 1, PA has been taken as an example) could be used as diagnostic biomarkers with high sensitivity and specificity for toxicity induced by TCM and its active ingredients, and they have the huge potential to be further applied in clinic.

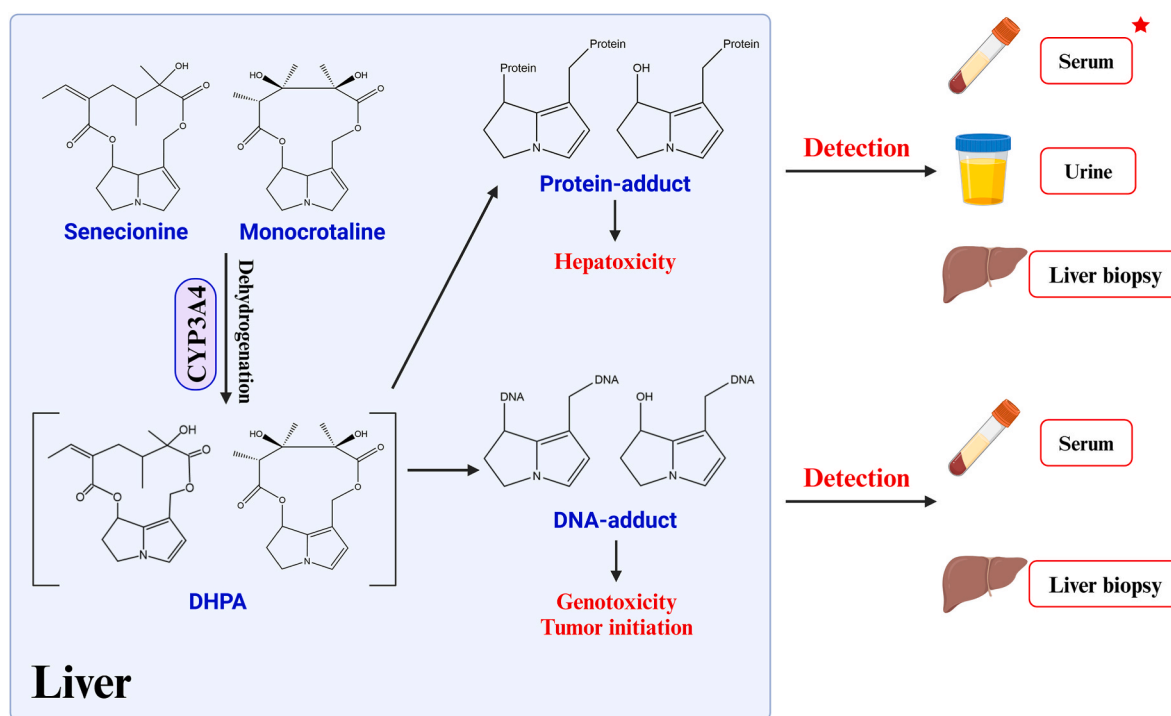
## 4. Non-coding RNAs as biomarkers for toxicity induced by TCM

Non-coding RNAs, mainly including microRNA (miRNA) and long non-coding RNA (lncRNA), have increasingly been applied in the study of toxicological biomarkers due to their high throughput, high sensitivity, and easy to quantify (Joseph, 2017). It has been reported that some miRNAs have already been used for the diagnosis and prognosis of cancer and cardiovascular diseases (Lujambio and Lowe, 2012; Wiczorek and Reszka, 2018; Zhou et al., 2018). Hence, non-coding RNAs hold significant potential as biomarkers for toxicity induced by TCM.

**Table 2**  
Protein/DNA-Adducts as the biomarker for toxicity induced by TCM.

| Type           | Natural product        | Adduct structure                                                                    | Reference                                                                                                                                                                                                                                                    |
|----------------|------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protein-adduct | Pyrrolizidine alkaloid |    | <a href="#">Ma et al. (2018)</a> ; <a href="#">Lin et al. (2011)</a> ; <a href="#">Gao et al. (2015)</a> ; <a href="#">Zhuge et al. (2019)</a> ; <a href="#">Ma et al. (2021a)</a> ; <a href="#">Ma et al. (2021b)</a> ; <a href="#">Yang et al. (2017a)</a> |
|                | Toosendanin            |    |                                                                                                                                                                                                                                                              |
|                | Diosbulbin-B           |   |                                                                                                                                                                                                                                                              |
| Estragole      |                        |  | <a href="#">Bergau et al. (2021)</a>                                                                                                                                                                                                                         |
| Geniposide     |                        |  | <a href="#">Li et al. (2019)</a>                                                                                                                                                                                                                             |
| DNA-adduct     | Aristolochic acid      |  | <a href="#">Jelakovic et al. (2012)</a> ; <a href="#">Stiborova et al. (2017)</a> ; <a href="#">Anger et al. (2020)</a>                                                                                                                                      |
|                | Pyrrolizidine alkaloid |  |                                                                                                                                                                                                                                                              |
|                | Estragole              |  |                                                                                                                                                                                                                                                              |





★ This method has been used in Nanjing criteria

Fig. 1. Diagram of the detection of DHPA-DNA/protein adducts *in vivo*.

#### 4.1. miRNA as the biomarker

miRNA is a class of endogenously short and highly conserved non-coding RNA composed of 18–25 nucleotides and displays a critical role in miscellaneous physiological and pathological processes including development, proliferation, differentiation, apoptosis, and energy metabolism (Bartel, 2004). miRNA has a good stability in the blood and can be easily detected with the development of miRNA detection technology, thereby endowing miRNA with the huge potential to be an emerging and minimally invasive biomarker for the diagnosis of TCM-induced toxicity (Bartel, 2004; Chen et al., 2008).

Previous research manifested that miRNA-122-3p and miRNA-194-5p were significantly up-regulated in the serum of mice given with *Dioscorea bulbifera* L. or its primary hepatotoxic compound diosbulbin-B. Notably, this increase occurred earlier than the change of ALT or AST, which are two classic biochemical indicators commonly used to assess liver injury (Yang et al., 2017b). Additionally, serum miRNA-5099 was significantly increased in mice treated with *Dioscorea bulbifera* L. or diosbulbin-B, but not in mice given with three other hepatotoxins, including acetaminophen, monocrotaline, and TSN (Yang et al., 2017b). Except finding that the level of miRNA-122-3p was up-regulated in serum from mice treated with TSN, diosbulbin-B, acetaminophen and monocrotaline, miRNA-367-3p was decreased only in mice exposed to TSN (Yang et al., 2019). Previous reports have already demonstrated the potential application of miRNA-122, which is highly abundant in hepatocytes, as a biomarker for the diagnosis of multifarious liver diseases such as patients with DILI (Su et al., 2012; Russo et al., 2017; Marin-Romero et al., 2020). However, the disadvantage of miRNA-122 is that only miRNA-122 could not distinguish the concrete type of liver disease. These above evidences further supports the usage of miRNA-122 as a universally sensitive biomarker for reflecting hepatic damage, but miRNA-5099 may serve as a specific biomarker for hepatotoxicity caused by *Dioscorea bulbifera* L., while miRNA-367-3p could be a specific biomarker for hepatotoxicity caused by TSN.

In nephrotoxicity induced by aristolochic acid I (a typical AA), miRNA-21-3p was markedly enhanced in both plasma and kidney, but not in non-target organs, and this increase was highly correlated with the dosage of AA (Pu et al., 2017). Notably, this upregulation occurred earlier than the elevation of traditional renal injury markers such as creatinine and BUN, as well as the onset of renal tubular injury (Pu et al., 2017). These finding suggest that miRNA-21-3p has the potential to be a sensitive early-warning biomarker for nephrotoxicity induced by AA and AA-containing herbal medicines. A previous study has manifested that miRNA-21 was generally associated with drug-induced kidney injury (DIKI), and the occurrence of DIKI can be predicted by detecting miRNA-21 in urine (Huang et al., 2020). As determining miRNA-21 in urine is easier than in plasma because urine-based detection is non-invasive, making it a more practical approach for clinical monitoring AA-induced nephrotoxicity. However, whether detecting miRNA-21 in urine can reliably predict nephrotoxicity specifically caused by AA needs further investigation.

Oxalic acid, primarily present in *Averrhoa carambola* L., has been reported to induce nephrotoxicity (Shihana et al., 2020). A previous study demonstrated that five microRNAs (miRNA-20a, miRNA-92a, miRNA-93, miRNA-195, miRNA-451) markedly decreased in acute renal damage patients intoxicated with oxalic acid as compared to patients with no injury and showed a markedly positive correlation with normalized urinary albumin, indicating the potential application of those miRNAs as biomarkers for reflecting its nephrotoxicity (Shihana et al., 2020). Despite this promising finding, further experimental studies and clinical investigations are needed to validate the potential application of these miRNAs for assessing nephrotoxicity induced by oxalic acid or *Averrhoa carambola* L.

Triptolide, a toxic compound from *Tripterygium wilfordii* Hook. f., is well-documented for its multiple organ toxicities, including hepatotoxicity (Kong et al., 2015), nephrotoxicity (Sun et al., 2013), and cardiotoxicity (Zhou et al., 2015). A study manifested that triptolide caused serious cardiotoxicity in rats, and characterized by significant

alternations in miRNA expression profiles (Wang et al., 2019). A total of 15 miRNAs were up-regulated in heart and plasma in rats treated with triptolide. Among those miRNAs, miRNA-615 and miRNA-127-5p reached the peak at the early stage, while the level of miRNA-15a-3p and miRNA-483-3p increased gradually and reached the peak at the late stage (Wang et al., 2019). These above four miRNAs hold potential as early-warning biomarkers for triptolide-induced cardiotoxicity, providing valuable insights into its toxic mechanism and offering the approach for timely intervention.

#### 4.2. LncRNA as the biomarker

LncRNA is defined as endogenous non-coding RNA whose transcript is more than 200 nucleotides (Bridges et al., 2021), and it participates in many biological processes in the body (Batista and Chang, 2013; Rinn et al., 2007). Some previous studies displayed that lncRNA-ENST00000446135 and lncRNA-ENST00000414355 might be novel biomarkers for reflecting cadmium-induced toxicity (Zhou et al., 2020; Moawad et al., 2021). Meanwhile, some researches pointed out the possibility of lncRNA as innovative toxicological biomarkers for toxicity induced by a variety of chemicals including some drugs like doxorubicin (Dempsey and Cui, 2017; Fa et al., 2021). In the liver damage induced by tripterygium glycosides, results from transcriptomics have revealed significant alternations in the expression of over 400 lncRNAs, but whether these lncRNAs can be used as toxicological biomarkers requires further experimental validation (An et al., 2023). Currently, there is still no much research about the participation of lncRNA in toxicity induced by TCM. More attentions should be directed towards uncovering the potential application of lncRNAs as biomarkers for toxicity induced by TCM, as this represents a promising yet underexplored area of research.

All above researches have illustrated that miRNA and lncRNA showed a huge potential to be diagnostic biomarkers for toxicity induced by TCM, because they are readily measurable in serum and with the advantage of high sensitivity and specificity. Moreover, previous studies have already evidenced that some miRNAs showed the huge potential to be used as early biomarkers to reflect toxicity induced by TCM (Table 3). Although the study on miRNA as biomarkers for toxicity has made some progressions, it is still of necessity that more clinical data and experiments are needed for using miRNAs as early-warning biomarkers.

#### 5. Endogenous metabolites as biomarkers for toxicity induced by TCM

Endogenous metabolites are a collection of small molecular compounds produced during metabolic processes in the whole body in response to external stimuli, and they can reflect the occurrence of disease or directly participate in its progression (Wishart, 2019; Qiu et al., 2023). Metabolomics, as a powerful analytical technique, is often used to analyze and detect the change of endogenous metabolites in body fluids or tissue samples. With its rapid analysis, high sensitivity and high throughput, metabolomics has become a valuable tool for discovering new biomarkers for toxicity (Ni et al., 2014; Wishart, 2019). It has already been widely used to find the toxic biomarkers of TCM.

A total of 32 types of endogenous metabolites were found in the serum from rats co-treated with a non-toxic dose of LPS and *Polygonum multiflorum* radix (PMR), which is the dried root tuber of *Polygonum multiflorum* Thunb., as well as its main compound 2,3,5,4'-tetrahydroxydiphenylethylene (TSG). Among these metabolites, ten biomarkers were associated with TSG-induced hepatic damage, including reduced levels of L-valine and L-proline, and increased levels of urea, caproic acid, among others. Notably, six of these biomarkers were also found to be related to the PMR-induced hepatotoxicity (Table 4), suggesting that these serum metabolites could serve as potential biomarkers for PMR-induced liver injury (Lin et al., 2021). By comparing the differential

**Table 3**

MicroRNA (miRNA) as the biomarker for toxicity induced by TCM.

| Type of toxicity | Toxic TCM (compounds)     | miRNA        | Trend | Reference             |
|------------------|---------------------------|--------------|-------|-----------------------|
| Hepatotoxicity   | DBL (Diosbulbin B)        | miRNA-122-3p | ↑     | Yang et al., 2017b    |
|                  |                           | miRNA-194-5p | ↑     |                       |
|                  |                           | miRNA-5099   | ↑     |                       |
|                  | MTSZ (Toosendanin)        | miRNA-122-3p | ↑     | Yang et al. (2019)    |
|                  |                           | miRNA-367-3p | ↓     |                       |
|                  |                           | miRNA-21-3p  | ↑     |                       |
| Nephrotoxicity   | ARI (Aristolochic acid I) | miRNA-20a    | ↓     | Pu et al. (2017)      |
|                  |                           | miRNA-92a    | ↓     |                       |
|                  |                           | miRNA-93     | ↓     |                       |
|                  | AC (Oxalic acid)          | miRNA-195    | ↓     | Shihana et al. (2020) |
|                  |                           | miRNA-451    | ↓     |                       |
|                  |                           | miRNA-15a-3p | ↑     |                       |
| Cardiotoxicity   | TWHF (Triptolide)         | miRNA-127-5p | ↑     | Wang et al. (2019)    |
|                  |                           | miRNA-483-3p | ↑     |                       |
|                  |                           | miRNA-615    | ↑     |                       |
|                  |                           |              |       |                       |

*Aristolochia* species, ARI; *Averrhoa carambola*, AC; *Dioscorea bulbifera* L., DBL; *Melia toosendan* Sieb. et Zucc., MTSZ; *Tripterygium wilfordii* Hook.f., TWHF.

metabolites in plasma samples obtained from PMR-induced liver injury and PMR-tolerance patients, 8 potential biomarkers were discovered (Table 4) (Hu et al., 2023). Among these metabolites, lysoPC (16:0), hypoxanthine, and taurochenodesoxycholic acid demonstrated the high sensitivity and accuracy in distinguishing between these two conditions, highlighting their potential clinical relevance (Hu et al., 2023).

*Veratrum nigrum* L. has been found to induce severe hepatic injury in mice, and subsequent analysis using UHPLC-Q-TOF-MS identified 15 endogenous metabolites in the liver (Table 4). These metabolites were proposed as potential biomarkers for hepatic injury caused by *Veratrum nigrum* L. (Wei et al., 2017). Another metabolomics study conducted in human hepatocytes identified 8 metabolites, including sphinganine, glutathione, l-palmitoylcarnitine, elaidic carnitine and so on (Table 4), as potential biomarkers for hepatic injury induced by emodin, triptolide, and AA *in vitro* (Liu et al., 2015). Meanwhile, Liu et al. reported that *Dysosma Versipellis*, a traditional Chinese medicine for eliminating phlegm, could cause severe liver damage in rats. In their study, 17 chemical compounds that significantly changed were preliminarily identified based on UHPLC-Q-TOF-MS analysis, including 5-hydroxyisourate, fructoseglycine, and so on. These findings suggest that they could be potential biomarkers for toxicity induced by *Dysosma Versipellis* (Liu et al., 2020).

Interestingly, hypoxanthine and lysoPC (18:1) exhibited significant changes in hepatotoxicity induced by both *Polygonum multiflorum* Thunb. and *Veratrum nigrum* L., but they showed the completely opposite direction. Similarly, sphinganine was found to be changed in hepatotoxicity induced by *Veratrum nigrum* L. and hepatic injury induced by emodin, triptolide, and AA *in vitro*, and lysoPC (20:3) was reported to be changed in hepatotoxicity induced by both *Veratrum nigrum* L. and *Dysosma Versipellis*. However, their trend of change was also completely opposite in these cases. Hypoxanthine, an intermediate product of purine metabolism, has not yet been reported as a potential biomarker for toxicity. On the other hand, sphinganine, an intermediate metabolite of sphingolipid synthesis, has already been identified as a potential

**Table 4**  
Endogenous metabolites as biomarkers for toxicity induced by TCM.

| Type of toxicity | Toxic TCM (Compound)                                    | Metabolite                    | Trend | Reference         |
|------------------|---------------------------------------------------------|-------------------------------|-------|-------------------|
| Hepatotoxicity   | PMR                                                     | Choline                       | ↓     | Hu et al. (2023)  |
|                  |                                                         | Deoxycholic acid              | ↓     |                   |
|                  |                                                         | D-Glutamic acid               | ↓     |                   |
|                  |                                                         | Hypoxanthine                  | ↑     |                   |
|                  |                                                         | LysoPC (16:0)                 | ↑     |                   |
|                  |                                                         | LysoPC (18:1)                 | ↓     |                   |
|                  | TSG                                                     | Phenylalanylphenylalanine     | ↓     | Lin et al. (2021) |
|                  |                                                         | Taurochenodesoxycholic acid   | ↑     |                   |
|                  |                                                         | 3-hydroxybutyric acid         | ↑     |                   |
|                  |                                                         | Caproic acid                  | ↑     |                   |
|                  |                                                         | D-galactose                   | ↑     |                   |
|                  |                                                         | DL-malic acid                 | ↑     |                   |
|                  |                                                         | D-mannose                     | ↑     |                   |
|                  |                                                         | Hexadecanoic acid             | ↑     |                   |
|                  |                                                         | L-proline                     | ↓     |                   |
|                  |                                                         | L-valine                      | ↓     |                   |
|                  |                                                         | Octadecane                    | ↑     |                   |
|                  |                                                         | Urea                          | ↑     |                   |
|                  | VNL                                                     | Hexadecenoic acid             | ↑     | Wei et al. (2017) |
|                  |                                                         | Hypoxanthine                  | ↓     |                   |
|                  |                                                         | Inosine                       | ↑     |                   |
|                  |                                                         | LysoPC (16:1)                 | ↑     |                   |
|                  |                                                         | LysoPC (18:1)                 | ↑     |                   |
|                  |                                                         | LysoPC (20:2)                 | ↑     |                   |
|                  |                                                         | LysoPC (20:3)                 | ↑     |                   |
|                  |                                                         | Octadecatrienoic acid         | ↑     |                   |
|                  |                                                         | Octadecadienoic acid          | ↑     |                   |
|                  |                                                         | Palmitoylcarnitine            | ↑     |                   |
|                  |                                                         | Sphinganine                   | ↓     |                   |
|                  |                                                         | Taurine                       | ↓     |                   |
|                  |                                                         | Tryptophan                    | ↑     |                   |
|                  |                                                         | Uric acid                     | ↑     |                   |
|                  |                                                         | Uridine                       | ↑     |                   |
|                  | DV                                                      | 5-Hydroxyisourate             | ↑     | Liu et al. (2020) |
|                  |                                                         | Fructoseglycine               | ↓     |                   |
|                  |                                                         | Tridecanoylglycine            | ↓     |                   |
|                  |                                                         | DHAP (8:0)                    | ↓     |                   |
|                  |                                                         | 12-Tricosanol                 | ↑     |                   |
|                  |                                                         | Stearoylglycine               | ↓     |                   |
|                  |                                                         | dUDP                          | ↓     |                   |
|                  |                                                         | Arachidylcarnitine            | ↓     |                   |
|                  |                                                         | dUTP                          | ↓     |                   |
|                  |                                                         | LysoPC (20:3)                 | ↓     |                   |
|                  |                                                         | L-carnitine                   | ↓     |                   |
|                  |                                                         | Phenylalanine                 | ↑     |                   |
|                  |                                                         | D-Ribose5-Phosphate           | ↑     |                   |
|                  |                                                         | Estradiol-17beta3-sulfate     | ↑     |                   |
|                  |                                                         | LysoPC (18:0)                 | ↑     |                   |
|                  | VMG (Emodin) -ARI (Aristolochic acid)-TWHF (Triptolide) | Tryptophyl-Proline            | ↓     | Liu et al. (2015) |
|                  |                                                         | Histidinyl-Phenylalanine      | ↓     |                   |
|                  |                                                         | C16 sphinganine               | ↑     |                   |
|                  |                                                         | Elaidic carnitine             | ↑     |                   |
|                  |                                                         | Glu Gly                       | ↑     |                   |
|                  |                                                         | Glutathione                   | ↓     |                   |
|                  |                                                         | L-Palmitoylcarnitine          | ↑     |                   |
|                  |                                                         | N1-Acetylspermidine           | ↑     |                   |
|                  |                                                         | N-Undecanoylglycine           | ↓     |                   |
|                  |                                                         | Sphinganine                   | ↑     |                   |
| Nephrotoxicity   | SNL                                                     | 19-hydroxydeoxycorticosterone | ↑     | Luo et al. (2018) |
|                  |                                                         | L-carnitine                   | ↓     |                   |
|                  | TWHF                                                    | LPE (18:2)                    | ↑     |                   |
|                  |                                                         | 3-indolepropionic acid        | ↓     |                   |
|                  |                                                         | LPE (20:1)                    | ↓     |                   |
|                  |                                                         | LysoPC (18:2)                 | ↑     |                   |
|                  |                                                         | LysoPC (20:5)                 | ↓     |                   |
|                  | SNL and TWHF                                            | LysoPC (22:5)                 | ↓     |                   |
|                  |                                                         | Alanine                       | ↓     |                   |
|                  |                                                         | Cholic acid                   | ↓     |                   |
|                  |                                                         | Isobutyryl-L-carnitine        | ↓     |                   |
|                  |                                                         | LysoPC (14:0)                 | ↓     |                   |
|                  |                                                         | LysoPC (16:1)                 | ↓     |                   |
|                  |                                                         | LysoPC (18:4)                 | ↓     |                   |
|                  |                                                         | LysoPC (20:2)                 | ↓     |                   |
|                  |                                                         | LysoPC (20:3)                 | ↓     |                   |

(continued on next page)

Table 4 (continued)

| Type of toxicity | Toxic TCM (Compound) | Metabolite         | Trend | Reference |
|------------------|----------------------|--------------------|-------|-----------|
|                  |                      | Palmitoylcarnitine | ↓     |           |
|                  |                      | Phenylalanine      | ↓     |           |
|                  |                      | Thymidine          | ↓     |           |
|                  |                      | Tryptophan         | ↓     |           |

*Aristolochia* species, ARI; *Dysosma Versipellis*, DV; *Polygonum multiflorum* Radix, PMR. *Strychnos nux-vomica* Linn., SNL; 2,3,5,4'-tetrahydroxyl-diphenylethylene, TSG; *Tripterygium wilfordii* Hook F., TWHF; *Ventilago maderaspatana* Gaertn.; VMG; *Veratrum nigrum* L., VNL.

biomarker of toxicity in ducks chronically exposed to fumonisin (Tran et al., 2006). LysoPC (18:1) and LysoPC (20:3), both classified as lyso-phosphatidylcholines, are key phospholipids that play diverse roles in a variety of biological processes. Alternations in lysoPC levels have already been found to be associated with liver injury (Shi et al., 2016; Furuhashi, 2020). Although their trends of change vary in different liver injuries, the observed alteration in hypoxanthine, sphinganine, lysoPC (18:1), and lysoPC (20:3) suggest that they may serve as potential biomarkers for the early detection and evaluation of hepatotoxicity.

In nephrotoxicity induced by *Strychnos nux-vomica* Linn. or *Tripterygium wilfordii* Hook F., plasma metabolic profiling analysis results identified 15 metabolites associated with *Strychnos nux-vomica* Linn. and 17 metabolites related to *Tripterygium wilfordii* Hook F. (Table 4) (Luo et al., 2018). Among these identified metabolites, 12 were found to be commonly altered in nephrotoxicity caused by both herbal medicines. These shared metabolites include alanine, cholic acid, and others, highlighting their critical role in the pathological processes of nephrotoxicity. These results suggest the huge potential of these metabolites as promising biomarkers for early detection and monitoring nephrotoxicity induced by *Strychnos nux-vomica* Linn. or *Tripterygium wilfordii* Hook F.

Metabolomics exhibits significant advantages in the qualitative and quantitative analysis of endogenous metabolites, but its ability to identify potential biomarkers for toxicity remains limited due to constraints in existing databases and reduced sensitivity in detection. Although metabolomics offers valuable insights and initial clues for discovering potential biomarkers, extensive clinical validations are required to confirm their applicability and reliability in clinical practice. Addressing these limitations through improved methodologies and robust databases will be helpful for enhancing the utility of metabolomics in discovering biomarkers for toxicity induced by TCM.

6. Endogenous proteins as biomarkers for toxicity induced by TCM

Some proteins serve as valuable biomarkers for toxicity because they are closely associated with the physiological functions and pathological states of the body. Changes in the expression, concentration, or activity of these proteins can sensitively and specifically reflect the toxic effects of drugs, providing critical insights into their impact on the body. Hence, these proteins hold significant potential as biomarkers for detecting, monitoring, and assessing toxicity induced by TCM.

Cytokines are soluble proteins secreted from a variety of cells with multiple biological functions. It is reasonable for considering cytokines as promising candidates for early-warning biomarkers due to their high susceptibility and ease of detection (Tu et al., 2021). It has been reported that mild immune stress could be one of the susceptibility-related factors of idiosyncratic DILI caused by *Polygoni multiflori* Radix (Boelsterli and Lee, 2014; Tu et al., 2019; Rao et al., 2021). A previous study revealed that 12 chemokines [interferon- $\gamma$ -inducible protein 10, monocyte chemotactic protein-1 (MCP-1), etc.] related to inflammation and immune regulation could be considered as potential biomarkers for identifying the susceptibility to hepatotoxicity caused by *Polygoni multiflori* Radix (Tu et al., 2019). Recently, results from two independent retrospective studies in clinic revealed that cytokines including tumor necrosis factor alpha (TNF $\alpha$ ), MCP-1 and vascular endothelial growth factor (VEGF) markedly increased in the serum from patients with

*Polygoni multiflori* Radix-induced DILI compared to *Polygoni multiflori* Radix-tolerant individuals. These results indicate that TNF $\alpha$ , MCP-1 and VEGF could serve as reliable biomarkers for predicting susceptibility to *Polygoni multiflori* Radix-related DILI (Tu et al., 2021). A previous study manifested that the altered expression of cluster of differentiation 30 ligand and interleukin-3 in the serum of mice could be potential biomarkers for liver injury caused by *Dioscorea bulbifera* L. (Sheng et al., 2014). These findings collectively underscore the potential of cytokines in predicting and diagnosing TCM-induced hepatotoxicity.

Kidney injury molecule-1 (KIM-1) is a type I cell membrane glycoprotein. Previous reports have demonstrated that KIM-1's expression was nearly undetectable under normal conditions but is markedly upregulated in renal cells following kidney damage (Ichimura et al., 1998, 2008; Griffin et al., 2019). Currently, KIM-1 has already been a novel and sensitive biomarker for acute kidney damage and been used as a biomarker in non-clinical and clinical drug development to help guide safety assessment by the Food and Drug Administration and European Medicines Agency (Bonventre et al., 2010; Dieterle et al., 2010; Abdel-salam et al., 2018). The increase of urine KIM-1 was earlier than classical indicators for kidney injury in AA-caused nephrotoxicity in mice, and KIM-1 could be detected with only 25  $\mu$ l of urine and was stable for at least 5 days (Sabbisetti et al., 2013). These results imply that KIM-1 may also serve as a valuable biomarker for identifying nephrotoxicity induced by AA. By using KIM-1 as a biomarker for renal injury, some previous studies demonstrated that esculetoside A from the root of *Phytolacca acinose* Roxb. could cause significant nephrotoxicity (Gu et al., 2024). Similarly, although the methanol extract of *Tetrorchidium didymostemon* leaves had no obvious effect on body weight, biochemical parameters, oxidative injury indices and histopathology, the increased KIM-1 level alerted people to its potential nephrotoxicity (Ebohon et al., 2020). These observations underscore the potential of KIM-1 as a promising biomarker for detecting and assessing nephrotoxicity induced by TCM or its toxic compounds.

In a previous study, various methods such as acute toxicity, cumulative toxicity, tolerance testing, and sub-chronic toxicity studies were adopted to evaluate the liver toxicity of *Sophorae tonkinensis* Radix et Rhizoma. The findings revealed that the level of serum cholinesterase (CHE) increased in a dose-dependent manner, while traditional liver injury biomarkers such as ALT and AST showed a downward trend in the sub-chronic toxicity study. This suggests that the elevation of serum CHE may serve as a potential auxiliary biomarker for evaluating liver injury induced by *Sophorae tonkinensis* Radix et Rhizoma (Wang et al., 2017). In a study of hepatotoxicity induced by Geniposide from *Gardeniae Fructus*, two biomarkers including glycine N-methyltransferase and glycogen phosphorylase were found to indicate hepatic damage significantly earlier than classic biomarkers (Wei et al., 2014). However, further experimental investigations and clinical evidences are required to ensure their reliability and accuracy in assessing hepatotoxicity.

Polyphyllins found in *Paridis Rhizoma*, including polyphyllin VI and polyphyllin VII, have been reported to induce hepatotoxicity by disrupting cholesterol biosynthesis, leading to hepatocyte apoptosis and lipid metabolism disorders (Li et al., 2022). Squalene synthase (SQLE) has been found to play a central role in regulating cholesterol biosynthesis, and polyphyllin I (another polyphyllin) was discovered to directly bind with SQLE. Consequently, SQLE is considered a potential biomarker for assessing liver injury induced by *Paridis Rhizoma* (Li



et al., 2023). However, despite these findings, clinical evidences regarding liver injury caused by *Paridis Rhizoma* remains limited, and the role of SQLE as a reliable biomarker has yet to be fully validated (Ding et al., 2021). Further in-depth studies are needed to confirm the clinical relevance of SQLE in liver injury and to elucidate the mechanisms underlying the hepatotoxic effects of polyphyllins.

Diosbulbin D, another toxic compound derived from *Dioscorea bulbifera* L., has been reported to undergo hepatic metabolism mediated by cytochrome P450 enzymes. This metabolic process produces a reactive metabolite capable of covalently binding to microsomal epoxide hydrolase, thereby triggering the formation of anti-microsomal epoxide hydrolase autoantibodies (Chen et al., 2011). Notably, a previous study has indicated that these autoantibodies can persist in the serum for a long period time (De Berardinis et al., 2000). Given their prolonged presence in the serum, anti-microsomal epoxide hydrolase autoantibodies have the potential to be promising biomarkers for assessing the hepatotoxicity of diosbulbin D. However, their clinical application requires further validation through comprehensive experimental studies and robust clinical evidences.

Studies have suggest that other proteins, such as beta 2-microglobulin, lipocalin-2, osteopontin, and cystatin C, may also serve as potential biomarkers for nephrotoxicity induced by traditional herbal medicines (Gao et al., 2023; Fuchs et al., 2014). These proteins listed above have been identified as potential biomarkers for assessing toxicity induced by TCM and its toxic compounds (Table 5). However, further more clinical evidences are needed to validate whether they can actually be used as biomarkers for TCM-induced toxicity.

**Table 5**  
Proteins as biomarkers for toxicity induced by TCM.

| Type           | Protein                                                        | Trend | Reference                                                        |
|----------------|----------------------------------------------------------------|-------|------------------------------------------------------------------|
| Cytokines      | Interleukin-1 beta                                             | ↑     | Tu et al. (2019); Tu et al. (2021)                               |
|                | Interleukin-6                                                  | ↑     |                                                                  |
|                | Interleukin-10                                                 | ↑     |                                                                  |
|                | Interferon gamma                                               | ↑     |                                                                  |
|                | Growth related oncogene alpha                                  | ↑     |                                                                  |
|                | Granulocyte-macrophage colony-stimulating factor               | ↑     | Sheng et al. (2014)                                              |
|                | Tumor necrosis factor alpha                                    | ↑     |                                                                  |
|                | Monocyte chemotactic protein-1                                 | ↑     |                                                                  |
|                | Monocyte chemotactic protein-3                                 | ↑     |                                                                  |
|                | Macrophage inflammatory protein-1 alpha                        | ↑     |                                                                  |
|                | Regulated upon activation normal T cell expressed and secreted | ↑     |                                                                  |
|                | Interferon gamma-inducible protein-10                          | ↑     |                                                                  |
|                | Vascular endothelial growth factor                             | ↑     |                                                                  |
|                | Cluster of differentiation 30 ligand                           | ↑     |                                                                  |
|                | Interleukin-3                                                  | ↓     |                                                                  |
|                | Kidney injury molecule-1                                       | ↑     |                                                                  |
| Other proteins | Glycogen phosphorylase                                         | ↑     | Sabbisetti et al. (2013); Gao et al., 2023; Ebohon et al. (2020) |
|                | Glycine N-methyltransferase                                    | ↑     |                                                                  |
|                | Cholinesterase                                                 | ↑     |                                                                  |
|                | Squalene synthase                                              | ↑     | Wang et al. (2017)                                               |
|                | Anti-microsomal epoxide hydrolase autoantibodies               | ↑     | Li et al. (2023)                                                 |
|                | Beta 2-microglobulin                                           | ↑     | Chen et al. (2011)                                               |
|                | Lipocalin-2                                                    | ↑     |                                                                  |
|                | Osteopontin                                                    | ↑     |                                                                  |
|                | Cystatin C                                                     | ↑     |                                                                  |

## 7. Conclusion

Herein, we have systematically discussed and summarized biomarkers for TCM-induced toxicity, and classified them in a logical and comprehensive manner. These biomarkers are characterized by their high sensitivity and specificity, making them suitable for be applied in clinic to reflect toxic damage caused by TCM.

Firstly, we explored the significant potential of DNA/protein adducts as biomarkers for TCM-induced toxicity. These adducts showed a high level of injury specificity and could accurately determine the type of damage induced by TCM. For example, pyrrole-protein adducts could indicate liver injury caused by TCM containing PA (Lin et al., 2011), while aristolactam-DNA adducts could reflect kidney damage induced by TCM containing AA (Jelakovic et al., 2012). However, there are still some shortcomings for DNA/protein adducts to be used as biomarkers, because they require precise instruments such as LC/MS for detection, resulting in the high testing cost. Furthermore, the measurement process is also susceptible to various factors, such as medication duration and potential interferences from other types of adducts. Additionally, the correlation between adducts and TCM-induced toxicity needs further validation through extensive clinical trials to ensure reliability and applicability.

Non-coding RNAs have emerged as potential early-warning biomarkers for TCM's toxicity due to their high sensitivity and tissue specificity. For instance, miRNA-122-3p has been reported to increase rapidly in liver injury induced by TCM containing diosbulbin-B, and its elevation occurs earlier than the change of classic biomarkers such as ALT and AST. Similarly, serum miRNA-5099 has been experimentally discovered as a specific biomarker for damage caused by diosbulbin-B (Yang et al., 2017b). Meanwhile, circulating non-coding RNAs can be detected using various RNA sequencing methods, which is more convenient. However, the current understanding of the biological functions and molecular mechanisms of non-coding RNAs remains limited. Therefore, it is necessary to clarify the importance of differential expression of non-coding RNAs in TCM's toxic reactions. What also cannot be ignored is that more effective, accurate, and feasible RNA detection methods should be developed to meet the requirements of medical diagnosis and to improve the early detection and management of TCM-induced toxicity.

Metabolites have the distinct advantage of serving as biomarkers because they can reflect pathological alterations in tissues or organs, and the metabolic characteristics of diseases can assess risk or provide specific disease subtypes. However, there are significant challenges that hinder the application of metabolites as new biomarkers for TCM-induced toxicity. The primary issue is the difficulty in the identification and validation of metabolites, largely due to the current limitations in metabolomics technology and the coverage of metabolomics is still restricted. On the other hand, metabolic changes are so complex, and currently there is no unified strategy or comprehensive analysis platform to analyze all metabolites of the entire phenotype. To address these limitations, it is essential to establish unified, standardized and universally recognized protocols. Considering the complexity and difficulty of high-throughput analysis, data integration and individual differences in metabolites, further research is required to fully realize the potential of metabolites as reliable biomarkers for assessing TCM-induced toxicity.

Other potential biomarkers, such as cytokines, KIM-1, and others, are also considered to exhibit high sensitivity as compared with classic biomarkers, offering promising prospects for further development as biomarkers for assessing toxicity induced by TCM. However, the biomarkers reviewed above still have their own shortcomings. It is evident that the identification and validation of novel early-warning biomarkers require more extensive experimental research and robust clinical data. Looking ahead, the development of appropriate combination diagnostic strategies and integrating multiple biomarkers for joint diagnosis may be the main development direction. For example, highly specific DNA/protein adducts could be applied to determine the type of damage

caused by TCM, while the highly sensitive non-coding RNAs could be assisted to indicate the severity of injury.

TCM exhibits unique advantages, including widespread applications, high efficiency, and safe in usage. However, we also need to pay more attentions to the potential toxicity associated with TCM. Traditional toxicological biomarkers often lack specificity, so they are insufficient to effectively and quickly indicate the complex toxicities caused by TCM. Therefore, it is necessary to seek new, sensitive, and early-warning biomarkers capable of accurately reflecting TCM-induced toxicity. Such advancement would significantly contribute to the rational and safe clinical application of TCM. In the future, more suitable and specific biomarkers are expected to be developed and applied in clinical practice with further in-depth studies, thereby ensuring both the efficacy and safety of TCM in clinic.

## CRediT authorship contribution statement

**Xinnan Gu:** Writing – original draft, Investigation, Data curation. **Yu Zou:** Investigation, Data curation. **Zhenlin Huang:** Investigation, Funding acquisition. **Mengjuan Wei:** Writing – review & editing. **Lili Ji:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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